2F Post-Tension Slab: Calculation Note vs Design Drawing

REVIEW AID - INDICATIVE. Not an engineer-approved check. Frame mapping inferred (see C). Status values: MATCH / DIFFERS / NOT IN DRAWING / NOT IN CALC / NOT CHECKED.

1 Answer in brief

- The calculation note (ADAPT-Builder, ACI 318, one frame 1-1D-B, 9.50 m span, 2019) and the drawing (Success Design & Build, Rev 01, 22-Aug-2023) agree on materials, strand type and the load intensities. They do NOT agree on the member modelled, the PT quantity, or the tendon geometry.
- I reproduced the note's own results from its section data (midspan tension 25.0 kg/cm2, Mu 110.75 T.m, balanced load 4.89 T/m = 73% of dead load). The note is internally consistent.
- The note designs a 215 cm wide x 45 cm deep band (T-section) over the full 9.5 m span. The drawing shows a 0.25 m slab between the drop panels in the D-F panel and no band beam there. On a slab-only section the note's own PT gives tension about 29.1 kg/cm2 (limit 28.6) and phiMn about 55 T.m against Mu about 101 T.m.
- The drawing provides 16 strands (173 T at 10.8 T) in the 5.95 m frame window at the F-line columns (max 18 anywhere); the note needs about 20 strands (213 T). Drawing profiles give about 3.2 T/m balanced load against the note's 4.89 T/m.
- Conclusion for the crack question: there is a documented mismatch between what was calculated and what is drawn. This is a plausible design-side contributor, not proof of cause. It needs the as-built section (was the 0.45 m band built?) and the stressing records to confirm.

2 Comparison table

#	Item	Calculation note	Drawing / my data	Status
1	Concrete	f'c 320 kg/cm2 (31.4 MPa), f'ci 240 (23.5 MPa), Ec 275 000 (27.0 GPa)	Drawing notes (C32 assumed in my screening)	MATCH
2	Strand	fpu 18 600 kg/cm2 (1824 MPa), Aps 0.99 cm2	ASTM A416 Gr.270 (assumed basis)	MATCH
3	Effective stress	Fse 10.91 T/cm2 = 1070 MPa, Pe 105.9 kN/strand, 10.80 T	Assumed 112.9 kN/strand in my base run	DIFFERS (assumption now replaced by note)
4	Loads	SW 4.602, SDL 1.487+0.630 line, LL 2.975 T/m	SDL 2.45 kPa, LL 4.90 kPa, SW 6.25 kPa; 0.63 T/m line and 3.75 T point load are not on the drawing	MATCH (SDL/LL); line load NOT IN DRAWING
5	Member modelled	T-section: web 215 x 45 cm, flange 595 x 25 cm, A 19 175 cm2	D-F panel: 0.25 m slab between drop panels; no band beam (note PTS. 0.25 M.THK.)	DIFFERS
6	Span / frame width	9.50 m, 5.95 m frame	D-F = 9.5 m; F-line column tributary about 54 m2	DIFFERS (width)
7	Force in frame	P = 213 T (about 20 strands)	16 strands = 173 T at x=19 and x=28; max 18 anywhere	DIFFERS
8	Balanced load	4.89 T/m (73%% of DL), drape 32.5 - 6.6 cm	3.2 T/m from sheet-16 profiles (drape of 8 / 12 / 31.7 cm per group)	DIFFERS
9	CGS limits	min bottom 5.6 cm, min top 5.2 cm; midspan 6.6 cm	Profile lows 4 cm in 24 of 32 marks; highs above 39.8 cm in 10 marks	DIFFERS

10	Columns	20 x 100 and 60 x 60	C1 600, C3 800, C4 1100 (square assumed)	DIFFERS
11	Punching	Joint 2 ratio 1.464 (V 87.22 T, d 41.23 cm)	My screening 1.52 - 1.55 at C1 (19 / 28, 24.3); joint 1 not reported in note	DIFFERS (both > 1.0)
12	Mild steel	Sections empty; unbalanced-moment steel zero	No mild reinforcement on drawing	NOT IN CALC / NOT IN DRAWING
13	Date / revision	2019, project 'MIH Office Building'	Rev 01, 22-Aug-2023	NOT CHECKED (same project / revision unconfirmed)
14	Frame identity	Frame 1-1D-B, no grid reference	Only 9.5 m span is D-F	INFERRED

3 Reproduction of the note and indicative re-checks

Reproduction (note data typed from the PDF; units as printed):

Quantity	Note	Reproduced
Section area / centroid	19 175 cm2 / 27.45 cm	19175 cm2 / 27.45 cm
Midspan bottom tension	25.01 kg/cm2	25.01 kg/cm2
Mu midspan (1.4D + 1.7L + secondary)	110.75 T.m	110.75 T.m
Balanced load	4.89 T/m (73%)	4.89 T/m (72.8%)
Strands for 213 T	-	19.7 (10.80 T each)

Re-check on the drawn section (hand method scaled from the note's moments; ACI-style, approximate):

Case	Net moment T.m	P/A kg/cm2	Midspan tension kg/cm2	Limit kg/cm2	Util
0.25 m slab, note PT (213 T)	26.9	14.3	29.1	28.6	1.02
0.25 m slab, drawing PT (16 strands, 3.2 T/m)	39.0	11.6	51.3	28.6	1.79

ULS flexure (Mu vs phiMn)	phiMn T.m	Mu T.m	Mu / phiMn
T-section 45 cm, 20 strands (note basis)	121	110.75	0.91
T-section 45 cm, 16 strands (drawing)	98	110.75	1.13
Slab 25 cm, 20 strands	55	101	1.83
Slab 25 cm, 16 strands	45	101	2.24

Supplementary mild steel on the slab-only section with 20 strands would be about 77 cm2 over the 5.95 m width (12.9 cm2/m), no mild steel is shown anywhere. Indicative only.

4 Effect on my Eurocode screening of using the note's basis

Variant	Pe kN	Service Case A fail (of 38)	Max util A	Service Case B fail	ULS A fail (of 13)	Punching fail (of 20)	Max punching
base (assumed Pe 112.9 kN)	112.9	10	1.63	22	7	10	2.08
calc-note Fse (Pe 105.9 kN)	105.9	12	1.75	22	7	10	2.10
calc-note Fse + line load 0.63 T/m (+1.04 kPa)	105.9	16	2.06	25	7	10	2.26

Using the note's lower Fse and adding the 0.63 T/m line load worsens every check. The line load is 'NOT IN DRAWING': confirm whether it exists on 2F.

5 Open items / questions for the designer

- Which grid frame is 1-1D-B? Confirm it is the D-F panel at x = 19 and 28.
- Was the 0.45 m x 2.15 m band built along the full 9.5 m span? If so, the drawing sheet is missing it; if not, the note does not cover the built section.
- Is the note (2019) the basis of Rev 01 (2023)? Revision history is not available.
- Where are the 0.63 T/m line load and 3.75 T point load located?
- Provide stressing records (elongation / gauge) to confirm the Fse of 1070 MPa was achieved.
- Mild reinforcement (top over supports, bottom, punching) is empty in the note and not drawn.
- Reconcile profile lows (4 cm) with the 5.6 cm minimum CGS used by the note, and the profile-length mismatches already logged (G24).

6 Limits of this check

The note was read from a printed PDF; numbers are typed from it and cited by section. Hand re-checks use ACI-style formulas to be comparable with the note and are not a Eurocode design check. Hyperstatic moments and the ADAPT/SAFE model were not reproduced. All results are indicative. Nothing is marked approved; engineer sign-off is a human step.